NeonPine – product risk assessment (rev 1, draft)

Product	NeonPine NP-1, **rev B** hardware (layout of 2026-10-06), firmware not yet written. Rev 1 of this document assessed rev A
Method	Hazard identification per EN ISO 12100 principles; risk levels per the EU GPSR / Safety Gate risk-assessment method (severity × probability). Electrical energy classified with the IEC 62368-1 ES/PS approach
Intended use	Decorative light for indoor, dry, domestic use, standing on a shelf or windowsill; supplied from a USB-C port or adapter; may run unattended overnight
Foreseeable misuse	Picked up while on and touched on the back; handled by children; placed on a wet windowsill; powered from a poor adapter; lamps bent or pulled; dropped
Users	General public incl. elderly people. Children are likely to be near it (not intended users: "not a toy, 14+")
Author / date	design team, 2026-10-05 (*to be reviewed and signed by the manufacturer*)

1. Energy sources (IEC 62368-1 style classification)

Source	Value	Class	Accessible?
USB VBUS / +5V	5 V, ≤ 0.5 A (PTC)	ES1 / PS1	yes (connector, back pads), acceptable
HV+ / SW node (C4, D1, D2, Q1, L11, R5, R21–R30, R12/C8)	**105 V DC** (123 V fault), boost can deliver well over 2 mA into a body (≈ 13 mA into 2 kΩ at 0.35 W)	**ES2** (above the 60 V DC / 2 mA ES1 limits)	rev A: yes. **Rev B: no.** All HVBUS copper is on the back, behind the screwed rear cover (basic safeguard); 0 HVBUS items on the front copper
Lamp leads (front, THT)	105 V behind 150 kΩ → **≤ 0.70 mA DC**	ES1 (current-limited)	yes, acceptable
C4 stored charge	≤ 5.5 mJ at 105 V; bleeds through 2.04 MΩ, τ ≈ 2 s (< 60 V after ≈ 1.1 s)	ES1 after 2 s	relevant only once the rear cover is opened
Input power	≤ 2.5 W (PTC hold limit at 5 V), typically 0.5 W	**PS1** (< 15 W)	no fire enclosure needed
Optical	orange neon, ≪ 60 lm, no UV/blue peak	expected **Exempt Group** (EN 62471)	yes
Thermal	< 1 W dissipated over 37 cm²	expected surface rise < 10 K	yes

ES2 parts must be separated from ordinary persons by at least a **basic safeguard**, such as an enclosure that can only be opened with a tool. Solder mask is **not** recognised as insulation. Firmware limits (duty clamp, OVP trip) are **not** counted as safeguards, because there is no IEC 60730 Class B software evaluation.

2. Risk matrix

Severity: **1** minor (no or slight injury) · **2** moderate (needs first aid or a doctor) · **3** serious (hospital or permanent) · **4** very serious / fatal. Probability over the product's life: **A** > 1/10 · **B** 1/100 · **C** 1/1 000 · **D** 1/10 000 · **E** < 1/100 000. Risk level: L low · M medium · H high · S serious (S and H are **not acceptable** for placing on the market).

Sev \ Prob	A	B	C	D	E
4	S	S	S	H	M
3	S	S	H	M	L
2	S	H	M	L	L
1	M	L	L	L	L

3. Hazard register

"Before" = rev A as designed. "After" = with the listed mitigations implemented and verified.

#	Hazard / scenario	Cause	Sev	Prob before	Risk before	Mitigation (design → protective → information)	Prob after	Risk after
H1	**Electric shock from the back**	105 V ES2 on bare SMD pads	2 (painful DC shock, startle, dropped glass)	A	**S**	**[B] Rear insulating cover** (FR-4 back plate or moulded cover) over the whole back,	E	L

#	Hazard / scenario	Cause	Sev	Prob before	Risk before	Mitigation (design → protective → information)	Prob after	Risk after
	side: the user picks the tree up while it is on and touches the back	(HV+, SW); no enclosure	– 3 for a child's wet hand			held by screws (tool needed), ≥ 0.4 mm solid insulation or ≥ 2 mm clearance from HV parts. Rating label on the cover		
H2	**Electric shock from the front HV+ tracks** after a scratch or mask defect	27 mm of HV+ track on F.Cu, covered by solder mask only	2	C	**M**	**[B] Route HV nets on the back only** (F.Cu keep-out for the HV class); front copper is then ES1 only	E	L
H3	Shock from the lamp leads	105 V on the anode side	1 (≤ 0.7 mA DC, below perception for most people)	B	L	150 kΩ on the **anode** side of every lamp (inherent); verify touch current ≤ 2 mA DC with the lamp shorted and with the transistor shorted	B	L
H4	Shock from C4 after unplugging, when the cover is opened for repair	residual charge	1	D	L	2.04 MΩ divider as a bleeder (τ ≈ 2 s); manual note: "no user-serviceable parts"	D	L
H5	Overvoltage, component overstress or fire after a firmware fault (PWM stuck on, ADC fault)	software	2	C	M	Hardware: D2 clamp (V_BR 111–123 V) < Q1 150 V, C4 250 V, D1 200 V, MMBTA42 300 V; R2 gate pull-down; F1 PTC; IWDG. **Verify by fault test:** force 60 % duty, open feedback (R7 shorted / R5 open), confirm D2 temperature and no flame	D	L
H6	Fire from a component failure (shorted Q1, shorted MLCC C1/C3, shorted D2)	single fault	3	D	M	PS1 source (USB ≤ 2.5 W through PTC); F1 0.5 A PTC; V-0 FR-4; parts > 5 mm from combustible packaging in use. **Single-fault tests** on Q1 D-S short, C3 short, D2 short, L11 short: no flame, PTC trips, surface < 90 °C	E	L
H7	Overheating, burns when touched	normal use	1	E	L	< 1 W total; confirm with a thermal test (ΔT < 10 K)	E	L
H8	**Cuts from broken glass**: lamp broken by a drop, a child, or bending the leads	glass envelope Ø4 mm on the front	2	B	**H**	Lamps lie flat and are **bonded with a dab of silicone** (item 6); rounded board edges (R 1.5 mm); stable stand (item 9); warnings: "Not a toy", "Do not bend the lamps", "If a lamp breaks, unplug and dispose of the product"	C	M → accept with warnings; consider an optional clear front guard for a "family" SKU
H9	**Choking / ingestion of small parts** by a small child (broken lamp, detached light sensor, glass fragments)	small parts, children nearby	3	C	H	Not a toy; mark "Not suitable for children under 3 years / keep out of reach of children" + 14+ recommendation; lamp bonding; light sensor seated flat; packaging warnings	D	M → accepted (a decoration, not a toy)

#	Hazard / scenario	Cause	Sev	Prob before	Risk before	Mitigation (design → protective → information)	Prob after	Risk after
H10	**Radioactive material** in neon lamps (Kr-85 "dark effect reduced" types)	supplier choice	2 (regulatory, negligible dose)	C	M	**[B] Supplier declaration that NEON-3 contains no radioactive additive**; incoming inspection; firmware handles the dark effect instead (keep-alive BAM, ignition kick)	E	L
H11	Hazardous substances (Pb, **Cd**, Hg, phthalates…)	**R8 GL5528 is a CdS photoresistor (cadmium)**; other parts, solder, cable	2	**A** (Cd is present by design)	**H** (regulatory: RoHS non-conformity)	**[B] Replace R8 with a cadmium-free phototransistor** (see 03-materials-compliance.md); RoHS + REACH supplier declarations; lead-free (SAC) assembly, ENIG; no Hg (neon, not fluorescent)	E	L
H12	Moisture or condensation on a windowsill → HV leakage, tracking	foreseeable misuse	2	C	M	Indoor dry use in the instructions; rear cover; HV clearance 0.5 mm functional; **post-assembly flux cleaning** (residues + 105 V → leakage, and drift of the 2 MΩ feedback)	D	L
H13	Unattended overnight use	intended use	3 (fire)	E	L	PS1, PTC, sleep timer and auto-off at daylight; no heat	E	L
H14	Unsafe or non-compliant USB adapter	external	3	D	M	Adapter not supplied; manual requires an adapter certified to IEC 62368-1 (CE / UL listed); device draws ≤ 0.15 A	D	M → information only, outside product scope
H15	Strangulation by the USB cable	small child	4	E	M	Cable not supplied (or ≤ 1 m if bundled); warning in the manual; not a toy	E	M → accepted with warning (common to all USB products)
H16	Sharp edges / points (star tips, PCB edge)	handling	1	C	L	All outline corners R 1.5 mm; routed + deburred edges; no V-score stubs on the product edge (panel tabs placed on the pot bottom)	D	L
H17	Tip-over → glass breakage (H8)	no stand	2	B	H	Cross-stand slot + stand piece (item 9), or a weighted base; stability test (10° incline)	D	L
H18	EMC: interference with radio/TV, or malfunction from ESD	boost converter, MCU clock; touch pads	1 (not safety-relevant)	–	–	See 02-emc-test-plan.md; worst case after ESD is a reset; HV is never higher than the setpoint after a reset (R2 keeps Q1 off)	–	–
H19	Photobiological (eye)	light	1	E	L	Orange neon, ≪ 60 lm; EN 62471 assessment → Exempt Group expected	E	L

4. Conclusion

Rev B status: the design mitigations for H1 (rear cover), H2 (HVBUS on the back only), H11 (Cd-free TEPT4400 sensor), H16/H17 (rounded edges, stand base) and H18 (masked touch pad, TVS) are implemented in hardware. The star touch pad was removed. Their effectiveness still has to be **verified on rev B samples** (§5) before sign-off. H10 (Kr-85 declaration) and the lamp-bonding process (H8) remain open.

Rev A conclusion (kept for the record): * **Rev A must not be placed on the market.** H1 is a **serious** risk and H8, H9 and H17 are **high** before mitigation. * H11 is **high** as a regulatory non-conformity (cadmium in the CdS LDR). * With the **[B]** actions (rear cover, HV routing on the back only, Kr-85-free lamp declaration, Cd-free light sensor) plus lamp bonding, the stand and the instructions, every residual risk is **L or M**. The M items are accepted for a decorative product with warnings (H8, H9, H14, H15). * Verification to be recorded in this document (§5) before the manufacturer signs it off.

5. Verification record (to fill in on rev B samples)

Test	Requirement	Result
Touch current, lamp leads, normal + lamp short + transistor C-E short	≤ 2 mA DC through 2 kΩ	
Accessibility probe (IEC 61032 test probe B) on the assembled product incl. cover	no ES2 part touchable	
Rear cover fixing needs a tool; cover pull / drop test	cover stays, no ES2 exposed	
C4 discharge after unplugging	< 60 V within 2 s	
Fault tests H5/H6 (duty stuck, open feedback, Q1/C3/D2/L11 short)	no fire, no part > 90 °C, PTC trips, no ES2 exposure	
Thermal test, 4 h at max brightness, 35 °C ambient	ΔT < 10 K on accessible surfaces	
Drop test 1 m onto hardwood (3 ×)	no exposed ES2; glass breakage noted	
Stability on a 10° incline	does not tip	
EN 62471 assessment	Exempt Group	
Supplier declarations (RoHS, REACH, no radioactive additive)	on file	

6. Revision history

Rev	Date	Change
1	2026-10-05	First issue on rev A layout
2	2026-10-06	Rev B: cover, HVBUS back-only, Cd-free sensor, stand, ESD measures implemented; verification pending